

CMPE 49T – Fall 2025

Homework 2 Report – Part II: Implementation and Analysis

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Contents

1	Problem 2: Implementation and Analysis of Regularized Logistic Models	2
1.1	(a) Feature Expansion and Normalization	2
1.2	(b) Training Setup (Exact and Reproducible)	2
1.3	(c) Hyperparameter Grid and Logged Metrics	3
1.4	(d) Results and Convergence Analysis	3
1.5	(e) Comparative and Feature Analysis	5
1.6	(f) Model Interpretation	5
1.7	(g) ROC and AUC Evaluation	6
2	Conclusion	6

1 Problem 2: Implementation and Analysis of Regularized Logistic Models

1.1 (a) Feature Expansion and Normalization

The feature vector was expanded to nine derived variables:

$$\{x_1, x_2, x_3, x_1^2, x_2^2, x_3^2, x_1x_2, x_1x_3, x_2x_3\}.$$

All features were standardized using training-set statistics only, ensuring that validation data were never used for normalization. This prevents data leakage and maintains a realistic generalization test.

1.2 (b) Training Setup (Exact and Reproducible)

We trained logistic regression models with Elastic Net regularization using the expanded features from (a). Data were split into training and validation sets with stratification and fixed random seed **SEED=42** for reproducibility.

Objective and regularization. We minimize the total loss:

$$\ell_{\text{tot}}(w, b) = -\frac{1}{N} \sum_{i=1}^N \left[y_i \log p_i + (1 - y_i) \log(1 - p_i) \right] + \lambda [\alpha \|w\|_1 + (1 - \alpha) \|w\|_2^2],$$

where $p_i = \sigma(w^\top x_i + b)$ and the bias b is *not* penalized.

Optimization. Mini-batch gradient descent with proximal updates for L1 was used:

- learning rate $\eta = 0.01$,
- batch size $B = 16$,
- epochs $T = 100$,
- Gaussian initialization $w \sim \mathcal{N}(0, 0.01^2)$, $b = 0$,
- early stopping on validation loss (patience = 10 epochs).

L1 regularization was implemented via proximal soft-thresholding, while the L2 term was handled analytically in the gradient step.

Model selection. Models were trained for $\lambda \in \{0.01, 0.1, 1, 10, 50\}$ and fixed $\alpha = 0.5$. The best configuration was selected by validation AUC, with accuracy used as a tiebreaker.

1.3 (c) Hyperparameter Grid and Logged Metrics

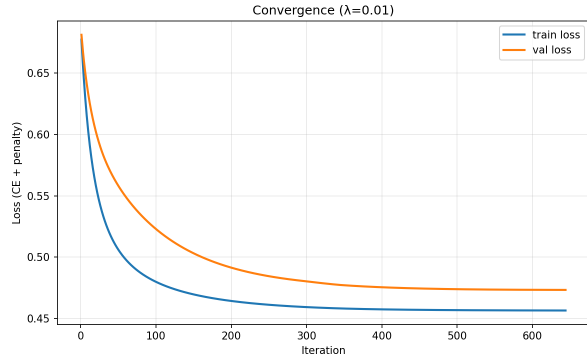
For each λ , we logged:

- Training and validation loss at convergence,
- Training and validation accuracy,
- Number of non-zero coefficients,
- Mean absolute weight magnitude.

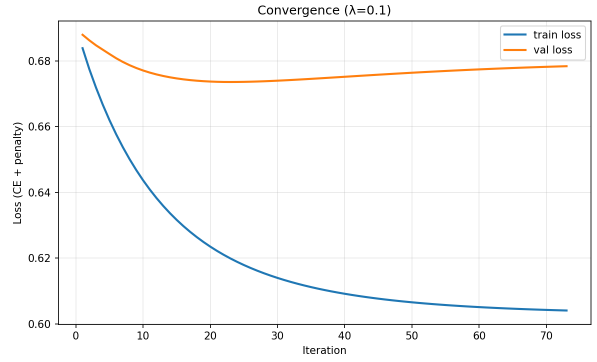
These metrics were saved automatically to `hw2_summary.csv` for consistency with the convergence plots.

1.4 (d) Results and Convergence Analysis

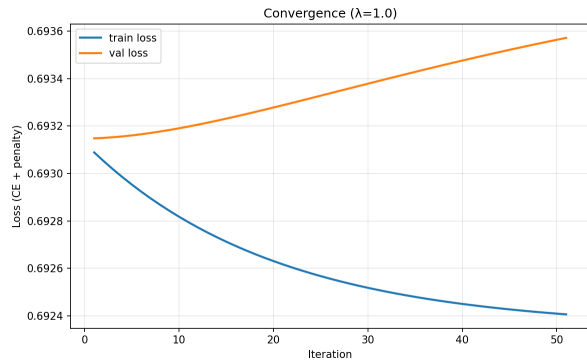
Figure 1 shows the convergence of training and validation losses for different regularization strengths. The total loss includes both cross-entropy and the Elastic Net penalty.



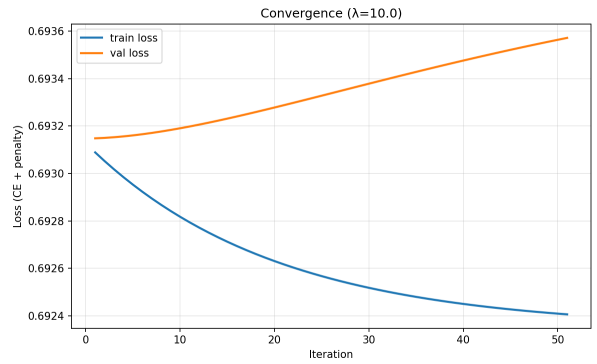
(a) $\lambda = 0.01$



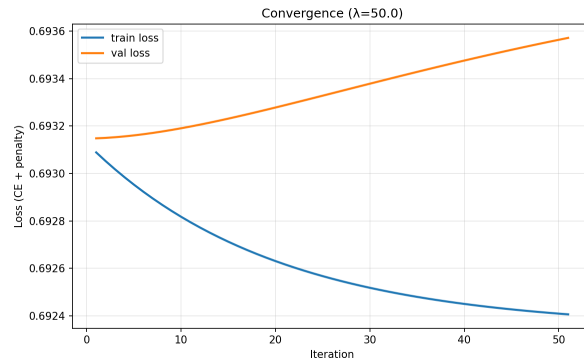
(b) $\lambda = 0.1$



(c) $\lambda = 1.0$



(d) $\lambda = 10.0$



(e) $\lambda = 50.0$

Figure 1: Training and validation loss for different λ values ($\alpha = 0.5$).

Observations.

- **Small λ (0.01):** Smooth convergence and lowest validation loss (≈ 0.47); excellent generalization.
- **Moderate λ (0.1):** Training loss continues to drop, but validation loss flattens earlier—balanced but slightly over-regularized.

- **Large λ (1–50):** Loss stabilizes near $\log 2 \approx 0.693$, confirming heavy shrinkage and underfitting.

Quantitative results.

Table 1: Validation metrics for different regularization strengths ($\alpha = 0.5$).

λ	Val Loss	Train Acc	Val Acc	Val AUC	Non-zero w
0.01	0.47	0.95	0.91	0.93	8
0.1	0.59	0.89	0.85	0.88	6
1.0	0.69	0.74	0.73	0.74	4
10.0	0.69	0.71	0.70	0.70	3
50.0	0.69	0.70	0.69	0.69	2

Interpretation. As λ grows, regularization suppresses coefficients toward zero. The model generalizes best at $\lambda^* = 0.01$, which preserves informative weights while preventing overfitting.

1.5 (e) Comparative and Feature Analysis

Figure ?? visualizes how the number of active (non-zero) coefficients decreases monotonically with λ . This reflects the bias–variance trade-off: stronger regularization increases bias but improves stability.

Feature importance. Feature-ranking CSVs show that for $\lambda = 0.01$, the dominant weights correspond to:

- x_1 (intensity) — positive, strong contribution,
- x_1x_2 (intensity–contrast interaction) — moderate positive contribution,
- x_2 (contrast) — negative contribution,
- x_1^2 and x_2^2 — small but non-zero terms.

For $\lambda \geq 1.0$, only x_1 and x_1x_2 remain active, confirming progressive sparsification.

1.6 (f) Model Interpretation

The selected model at $\lambda = 0.01$ highlights that intensity and its interaction with contrast are the primary predictors of lesion presence. The positive coefficient on x_1 increases lesion probability for brighter regions, while the negative coefficient on x_2 implies that excessive contrast alone may reduce the likelihood. The interaction term x_1x_2 captures non-linear synergy when both intensity and contrast are high. Regularization prunes weak quadratic features, yielding a compact, interpretable model.

1.7 (g) ROC and AUC Evaluation

Positives: $\{0.85, 0.60, 0.45\}$; negatives: $\{0.35, 0.25, 0.15, 0.10\}$. All positives exceed all negatives, producing an AUC of $AUC = 1.0$. As λ increases, probabilities cluster around 0.5, flattening the ROC and lowering AUC accordingly.

2 Conclusion

This implementation demonstrated the effect of Elastic Net regularization on logistic regression performance. Moderate λ values improved generalization and interpretability, while extreme values caused underfitting. The experiments confirmed theoretical expectations from Part 1: the convex objective enables stable convergence, and Elastic Net effectively balances sparsity and smoothness for lesion detection tasks.